

الإختصاص والرمز : المعلوماتية – فرع البرمجيات (1010)

المادة : لغة XML

المستندات المسموح بها : لا شيء

المعدل : (10)
المدة : ساعة ونصف

Treat all the following exercises:

I. (4 pts)

For the following document, find all mistakes and comment them. Indicate if they are XML validity mistakes or syntax and explain how to correct them.

```
1. <?xml version="1.0" encoding="iso-8859-1"?>
2. <!DOCTYPE mytypedoc [
3.
4. <!ENTITY copy "right">
5.
6. <!ELEMENT document (p*|(element,p)*)>
7. <!ELEMENT p (a|(a,b))*>
8. <!ELEMENT a (#PCDATA)>
9. <!ELEMENT b (#PCDATA)>
10. <!ELEMENT a (#PCDATA)>
11. <!ELEMENT element (#PCDATA)>
12.
13. <!ATTLIST document ii ID #REQUIRED>
14. <!ATTLIST document oo ID #REQUIRED>
15.
16. <!ATTLIST p att1 NMTOKEN #IMPLIED
17. att2 IDREFS #IMPLIED>
18.
19. ]>
20.
21.
22.
23. <document ii="ii" oo=aa>
24.
25. <element> Yo !</element>
26.
27. <p att1="c2de">
28. <b>ole</b>
29. <b>hop</b>
30. </p>
31.
32. <p att2="ii oo">
33. <a />
34. <a><b></a>
35. </p>
36.
37. %copy; myself & you.
38.
39. </document>
```

II. (3 Pts)

Examine this DTD that describes the content of the file catalog.xml, and answer the questions that follow:

```
<!ELEMENT catalog 0(caption, structure, work+)>
<!ELEMENT caption (#PCDATA)>
<!ELEMENT structure (title, author, type, date)>
<!ELEMENT work (title, author, type, date)>
<!ELEMENT title (#PCDATA)>
<!ELEMENT author (#PCDATA)>
<!ELEMENT type (#PCDATA)>
<!ELEMENT date (#PCDATA)>
```

- How many elements can contain the element « catalog», and which?
- How many « work » can contain a file « catalog.xml »?
- An element « title » can contain tagged content? Explain.

III. (3 pts)

Construct a valid XML document according to the external DTD given below. This describes information an estate agent might wish to store about a house.

```
<!ELEMENT houses (house*)>
<!ELEMENT house (room+)>
<!ELEMENT room (dimensions, windows, furniture*)>
<!ELEMENT dimensions (width, length)>
<!ELEMENT windows (window*)>
<!ELEMENT window (elevation, width, height)>
<!ELEMENT furniture (#PCDATA)>
<!ELEMENT width (#PCDATA)>
<!ELEMENT length (#PCDATA)>
<!ELEMENT height (#PCDATA)>
<!ELEMENT elevation EMPTY>
<!ATTLIST room name ID #REQUIRED
           colour IDREF #IMPLIED>
<!ATTLIST elevation facing (north | south | east | west) #REQUIRED>
```

IV. (5 pts)

Translate the following XML schema into a DTD:

```
<xsd:schema xmlns:xsd="http://www.w3.org/2001/XMLSchema"
  targetNamespace="http://www.journaux.com/">
  <xsd:element name="journal" type="JournalType"/>
  <xsd:complexType name="JournalType">
    <xsd:sequence>
      <xsd:element name="article" minOccurs="0" maxOccurs="unbounded">
        <xsd:complexType>
          <xsd:sequence>
            <xsd:element name="title" type="xsd:string"/>
            <xsd:element name="pages">
              <xsd:complexType>
                <xsd:element name="page" minOccurs="0" maxOccurs="unbounded">
                  <xsd:simpleType>
                    <xsd:restriction base="xsd:int">
                      <xsd:maxExclusive value="100"/>
                    </xsd:restriction>
                  </xsd:simpleType>
                </xsd:element>
              </xsd:complexType>
            </xsd:element>
            <xsd:element name="author" type="xsd:string"/>
            <xsd:element name="abstract" type="xsd:string"/>
          </xsd:sequence>
          <xsd:attribute name="href" type="xsd:string"/>
        </xsd:complexType>
      </xsd:element>
    </xsd:sequence>
    <xsd:attribute name="title" type="xsd:string"/>
  </xsd:complexType>
</xsd:schema>
```

V. (5 pts)

Given the following XML file:

```
<?xml version="1.0" encoding="ISO-8859-1" ?>
```

```
<recipe>
```

```
  <header>
```

```
    <author>Casimir</author>
```

```
    <title>Recipe of Gloubi-Boulga</title>
```

```
    <remark>For one person </remark>
```

```
  </header>
```

```
  <procedure>
```

```
    Fill a bowl with the strawberry jam, grated chocolate,  
    mashed bananas, hot mustard, and sausage from Toulouse  
    crushed raw but warm. Mix vigorously until a  
    light brown slurry. It is normal to have lumps. The  
    proportions are approximately equal for all the ingredients, but it  
    can vary according to personal tastes.
```

```
  </procedure>
```

```
</recipe>
```

Create a XSL style sheet allowing from this recipe card to produce a HTML page that:

- has as title the content of the tag <title>;
- starts by a title <h1> having as content the content of the tag < title>;
- gives then the name of the recipe author;
- displays then the word **Remark:** then the content of the tag <remark>;
- displays **Procedure** at level <h2> ;
- in a paragraph, presents the procedure to follow.